

Problem

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ be a linear transformation such that

$$T \underbrace{\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}}_{\vec{x}_1} = \begin{bmatrix} 4 \\ 4 \\ 0 \\ -2 \end{bmatrix} \text{ and } T \underbrace{\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix}}_{\vec{x}_2} = \begin{bmatrix} 4 \\ 5 \\ -1 \\ 5 \end{bmatrix}. \text{ Find } T \begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix}.$$

First, we need to see if $\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix}$ can be written as a linear combination of $\vec{x}_1$ and $\vec{x}_2$.

$$a_1 \vec{x}_1 + a_2 \vec{x}_2 = \begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix} \rightarrow a_1 \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} + a_2 \begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & 4 & -7 \\ 3 & 0 & 3 \\ 1 & 5 & -9 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 4 & -7 \\ 0 & -12 & 24 \\ 0 & 1 & -2 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 4 & -7 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array} \right]$$

$$\rightarrow a_1 = 1, a_2 = -2 \rightarrow \begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix} = (1) \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix}$$

Then we use property ③ of linear transformations:

③

$$\begin{aligned} T(\vec{y}) &= T(a_1 \vec{x}_1 + a_2 \vec{x}_2 + \cdots + a_k \vec{x}_k) \\ &= a_1 T(\vec{x}_1) + a_2 T(\vec{x}_2) + \cdots + a_k T(\vec{x}_k), \end{aligned}$$

$$\begin{aligned} T\left(\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix}\right) &= T\left((1) \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix}\right) = (1) T\left(\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}\right) - 2 T\left(\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix}\right) \\ &= \begin{bmatrix} 4 \\ 4 \\ 0 \\ -2 \end{bmatrix} - 2 \begin{bmatrix} 4 \\ 5 \\ -1 \\ 5 \end{bmatrix} \\ &= \begin{bmatrix} -4 \\ -6 \\ 2 \\ -12 \end{bmatrix} \end{aligned}$$

$$T\left(\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix}\right) = \begin{bmatrix} -4 \\ -6 \\ 2 \\ -12 \end{bmatrix}$$

Matrix Transformations

Theorem

Every matrix transformation is a linear transformation.

Consider the matrix transformation $T_A(\vec{x}) = A\vec{x}$. We will show that such a transformation is in fact linear, i.e., it preserves addition and scalar multiplication.

① Preservation of addition: ($A_{m \times n}$, $\vec{x}$ and $\vec{y} \in \mathbb{R}^n$, $A\vec{x} \in \mathbb{R}^m$)

$$\begin{aligned}T_A(\vec{x} + \vec{y}) &= A(\vec{x} + \vec{y}) = (x_1 + y_1)\vec{A}_1 + (x_2 + y_2)\vec{A}_2 + \cdots + (x_n + y_n)\vec{A}_n \\A(\vec{x} + \vec{y}) &= x_1\vec{A}_1 + x_2\vec{A}_2 + \cdots + x_n\vec{A}_n + y_1\vec{A}_1 + y_2\vec{A}_2 + \cdots + y_n\vec{A}_n \\A(\vec{x} + \vec{y}) &= A\vec{x} + A\vec{y} = T_A(\vec{x}) + T_A(\vec{y})\end{aligned}$$

② Preservation of scalar multiplication ($A_{m \times n}$, $c \in \mathbb{R}$, $\vec{x} \in \mathbb{R}^n$, $A\vec{x} \in \mathbb{R}^m$)

$$\begin{aligned}T_A(c\vec{x}) &= A(c\vec{x}) = (cx_1)\vec{A}_1 + (cx_2)\vec{A}_2 + \cdots + (cx_n)\vec{A}_n \\&= c(x_1\vec{A}_1 + x_2\vec{A}_2 + \cdots + x_n\vec{A}_n) \\&= c(A\vec{x}) = cT_A(\vec{x})\end{aligned}$$

Matrix Transformations

Question

Can you come up with an example of a transformation that is NOT a matrix transformation? Explain why it is not a matrix transformation.

As an example consider $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(\vec{x}) = \vec{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. You can easily check that T does not preserve $\vec{0}$ ($T(\vec{0}) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \neq \vec{0}$) and hence it is not a linear transformation.

Also, we cannot find a matrix A such that for all $\vec{x} \in \mathbb{R}^2$ we have $A\vec{x} = \vec{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Again consider $\vec{x} = \vec{0}$, then $\vec{0} = \vec{0} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ which is impossible.

As another example, consider $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(\vec{x}) = \begin{bmatrix} x^2 \\ y \end{bmatrix}$.

$$A\vec{x} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} a \\ c \end{bmatrix} + y \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} x^2 \\ y \end{bmatrix} \rightarrow \begin{cases} ax + by = x^2 \\ cx + dy = y \end{cases} \rightarrow \text{NOT possible}$$

Matrix and Linear Transformations

Theorem

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then T is a matrix transformation. Furthermore, T is induced by the **unique** matrix

$$A = \begin{bmatrix} T(\vec{e}_1) & T(\vec{e}_2) & \cdots & T(\vec{e}_n) \end{bmatrix},$$

where $\vec{e}_j$ is the j^{th} column of I_n , and $T(\vec{e}_j)$ is the j^{th} column of A .

Problem

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation defined by $T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 2y \\ x - y \end{bmatrix}$ for each $\vec{x} \in \mathbb{R}^2$. Find the matrix, A , of T .

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \vec{e}_1 & \vec{e}_2 \end{bmatrix} \quad T(\vec{e}_1) = T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \quad \quad T(\vec{e}_2) = T \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$A = \begin{bmatrix} T(\vec{e}_1) & T(\vec{e}_2) \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}$$

$$\text{Note that in fact } T(\vec{x}) = A\vec{x} = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 2y \\ x - y \end{bmatrix}.$$

Problem

Let $T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be transformations defined by $T_1 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} xy \\ x+y \end{bmatrix}$ and

$T_2 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ y \\ -x+2y \end{bmatrix}$ respectively. Determine whether these transformations are linear or not and (if possible) find the matrix corresponding to the transformation.

T_1 is NOT linear since it does not preserve negative of a vector.

$$\left. \begin{array}{l} T_1(\vec{-x}) = T_1 \begin{bmatrix} -x \\ -y \end{bmatrix} = \begin{bmatrix} xy \\ -x-y \end{bmatrix} \\ -T_1(\vec{x}) = \begin{bmatrix} -xy \\ -x-y \end{bmatrix} \end{array} \right\} \Rightarrow T_1(\vec{-x}) \neq -T_1(\vec{x})$$

T_2 is a linear transformation since

$$T_2(\vec{x}) = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ y \\ -x+2y \end{bmatrix}$$

Hence T_2 is a matrix transformation and every matrix transformation is a linear transformation.

$$T_2 = [\vec{e}_1 \quad \vec{e}_2] \quad T_2(\vec{e}_1) = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \quad T_2(\vec{e}_2) = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \quad A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ -1 & 2 \end{bmatrix}$$

One can also show that T_2 is a linear transformation by showing that it preserves vector addition and scalar multiplication.

$$\vec{v} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \quad \vec{u} = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$T_2(\vec{v} + \vec{u}) = \begin{bmatrix} 2(x_1 + x_2) \\ y_1 + y_2 \\ -(x_1 + x_2) + 2(y_1 + y_2) \end{bmatrix} = \begin{bmatrix} 2x_1 \\ y_1 \\ -x_1 + 2y_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ y_2 \\ -x_2 + 2y_2 \end{bmatrix} = T_2(\vec{v}) + T_2(\vec{u})$$

$$T(c\vec{v}) = \begin{bmatrix} cx_1 \\ cy_1 \\ -cx_1 + 2cy_1 \end{bmatrix} = c \begin{bmatrix} x_1 \\ y_1 \\ -x_1 + 2y_1 \end{bmatrix} = cT_2(\vec{v})$$

Note that the downside of this method is that it does not provide the matrix A .